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Electrosurgical devices with a single conductive tubular element for accessing anatomical structures

Abstract

An electrosurgical medical device may include a conductive tubular element disposed about an outer surface of an elongate tubular member at a distal portion of the tubular member. The conductive tubular member may be delivered to a treatment site of a gastrointestinal (GI) tract of a patient and electrically activated. While electrically activated, the conductive tubular element may create an opening in a wall of the GI tract and a pseudocyst. After the incision is made, the conductive tubular element may be withdrawn from the treatment site and a drainage device may be positioned in the incision for drainage of the pseudocyst.

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Background/Summary

(1) This application is a Divisional of co-pending U.S. Non-Provisional application Ser. No. 15/661,670, filed Jul. 27, 2017, which claims the benefit of U.S. Provisional Application No. 62/368,639, filed Jul. 29, 2016. The contents of U.S. Non-Provisional application Ser. No. 15/661,670 and U.S. Provisional Application No. 62/368,639 are incorporated by reference in their entirety.

TECHNICAL FIELD

(1) The present invention relates generally to medical devices, and more particularly to electrosurgical devices, systems, and related methods for using a tubular conductive element to electrosurgically create an opening for gaining access into an anatomical structure of a patient.

BACKGROUND

(2) A cystotome is an electrosurgical medical device that cuts tissue through application of electrical current in order to gain access into for example a pancreatic pseudocyst. Once the tissue is cut and access into the pseudocyst is obtained, a subsequent procedure may be performed on the pseudocyst, such as insertion of a stent into and/or draining of fluid from the pseudocyst. FIGS. 1A-1D show a side view of a distal portion **102** of an example cystotome **100** cutting tissue at a treatment site **104** within a patient to gain access into a pseudocyst **106**. An example type of pseudocyst **106** may be a pancreatic pseudocyst that pushes up against a wall **108** of the stomach or duodenum (e.g., a transgastric wall or a transduodenal wall) or of another area of the gastrointestinal (GI) tract, causing a bulge in the wall **108**.

(3) The cystotome **100** may include three elongate coaxial components: a conductive needle knife **110**, an inner catheter **112** disposed over the needle knife **110**, and an outer catheter **114** disposed over the inner catheter **112**. A ring-shaped conductive member **116** may be attached to a distal end **118** of the outer catheter **114**. The conductive member **116** may be referred to as a conductive distal tip **116** of the outer catheter **114**. The inner catheter **112** may function to electrically insulate the needle knife **110** from the conductive distal tip **116**. In a particular embodiment of the cystotome **100**, the needle knife **110** may have a diameter of 0.038 inches (0.9652 millimeters), the inner catheter **112** may have an outer diameter of 5 French (Fr) (1.667 millimeters), and the outer catheter **114** may have an outer diameter of 10 French (3.333 millimeters). A maximum diameter of the conductive distal tip **116** may also be or relatively close to 10 French.

(4) Although not shown in FIGS. 1A-1D, each of the needle knife **110**, the inner catheter **112**, and the outer catheter **114** may be operatively coupled to a handle assembly at a proximal portion. Through operation of the handle assembly, the needle knife **110**, the inner catheter **112**, and the outer catheter **114** may be configured to longitudinally move relative to each other, or may be

locked in fixed positions relative to each other. In addition, although not shown in FIGS. 1A-1D, the needle knife **110** and the distal tip **116** may be alternately connected to and electrically activated by a power source via the handle assembly.

(5) Referring to FIG. 1A, the distal portion **102** of the cystotome **100** may be delivered to the treatment site **104**. As shown in FIG. 1A, a distal portion of the needle knife **110** may be moved distally past a distal end **120** of the inner catheter **112** in order to be exposed to the outer surroundings and contact the tissue it is intended to cut. In addition, the distal tip **116** may be longitudinally positioned a distance proximally away from the distal end **120** of the inner catheter **112** so that is a sufficient distance away the exposed distal portion of the needle knife **110** to avoid any electrical shorting.

(6) Referring to FIG. 1B, the power source connected to the needle knife **110** may be activated, causing electrical current to flow through the needle knife **110**. The distal portion **102** may be distally advanced so that the distal portion of the electrically activated needle knife **110** contacts and cuts into the bulging part of the wall **108**. The distal portion **102** may continue to be distally advanced so that the needle knife **110** cuts into the wall **108** and then into the pseudocyst **106** to make an incision **122**. As shown in FIG. 1B, at this point during the procedure, distal portions of the needle knife **110** and the inner catheter **112** may have gained access to and be disposed in the pseudocyst **106**, while the outer catheter **114** may still be outside the pseudocyst **106**.

(7) Referring to FIG. 1C, the incision **122** made by the needle knife **110** may be too small for a subsequent procedure, such as insertion of a stent or other drainage device into the pseudocyst **106**. To make the incision **122** larger, the 10 French conductive distal tip **116** may be used. In particular, the needle knife **110** may be electrically deactivated and disconnected from the power source, and proximally withdrawn away from the treatment site **104** to outside of the patient. A wire guide **124** may then be distally advanced through the inner catheter **112** until a distal portion of the wire guide **124** is disposed in the pseudocyst **106**. The wire guide **124** may help to stabilize the inner catheter **112** in the pseudocyst **106** after the needle knife **110** is withdrawn. FIG. 1C shows the distal portion of wire guide **124** having curled within the pseudocyst **106**. In addition, the conductive distal tip **116** may be electrically coupled to the power source via a conductive wire **126** attached to the distal tip **116**. The conductive wire **126** may be attached and electrically connected to the distal tip **116** using solder **128**. Additionally, the wire **126** may longitudinally extend in between an outer surface of the inner catheter **112** and an inner surface of the outer catheter **114**. Although not shown, the wire **126** may proximally extend to the handle assembly outside of the patient, and the handle assembly may electrically couple the wire **126** to the power source. The outer catheter **114** may be distally advanced toward the wall **108**. The distal tip **116** may be electrically activated using the power source and may perform a second cut into the wall **108** and then into the pseudocyst **106** to enlarge the incision **122**. The inner catheter **112** may function as an insulator between the electrically activated distal tip **116** and the wire guide **124**.

(8) Referring to FIG. 1D, after the distal tip has sufficiently cut into the pseudocyst **106** and enlarged the incision **122**, the conductive distal tip **116** may be deactivated and the inner and outer catheters **112**, **114** may be proximally withdrawn from the treatment site **104** to outside the patient. The wire guide **124** may be left in place, with the distal portion of the wire guide **124** remaining in the pseudocyst **106**. Thereafter, a subsequent procedure associated with the pseudocyst **106** may be performed. For example, a delivery device may be distally advanced over the wire guide **124** toward the treatment site **104** to place a stent or other drainage device in the pseudocyst **106**.

(9) In some situations, insertion of a stent or other drainage device in a range from five French to ten French may be desirable. However, the size of the incision **122** after being enlarged by the ten French conductive tip **116** may be too large for such a stent or other drainage device to be securely maintained in the pseudocyst. At the same time, the size of the incision **122** made by the needle knife **110** may be too small. Merely reducing the diameter of the conductive tip **116** to be smaller than ten French may not be feasible since the power required to cut through the wall **108** and into

the pseudocyst **106** (e.g, 80-120 Watts) may cause such a reduced-size conductive tip to generate heat at a level that melts the solder attaching the conductive tip **116** to the wire **126**, which in turn may cause the distal tip **116** to lose its electrical connection with the wire **126** and ultimately the power source. Additionally, having to perform two cuts with two different conductive elements (i.e., the needle knife **110** and the conductive tip **116**) to form the incision **122** is time consuming and cumbersome. As such, it may be desirable for a cystotome to have a single conductive element to perform the cutting and make a sufficiently-sized incision that allows a five-to-ten French stent or other drainage device to be inserted into and securely maintained in the pseudocyst.

BRIEF SUMMARY

(10) By way of introduction, the below embodiments relate to electrosurgical devices, systems, and methods for using a tubular conductive element to electrosurgically create an opening for gaining access into an anatomical structure of a patient. In a first embodiment, an electrosurgical device includes an elongate tubular member, a conductive tubular element, and an elongate conductive member. The elongate tubular member extends from a proximal portion to a distal portion and includes a body and a wireguide lumen longitudinally extending in the body from the proximal portion to the distal portion. The body includes an insulating material. The conductive tubular element is affixed to the body and disposed about an outer surface of the body at the distal portion. The insulating material of the body insulates the wireguide lumen from the conductive tubular element. The elongate conductive member longitudinally extends adjacent the body from the proximal portion to the distal portion. A distal portion of the elongate conductive member is attached and electrically connected to the conductive tubular element.

(11) In some embodiments, the conductive tubular element is disposed over the distal portion of the elongate conductive member.

(12) In some embodiments, the conductive tubular element includes a conductive cannula.

(13) In some embodiments, the conductive tubular element includes a conductive coil.

(14) In some embodiments, the conductive coil and the elongate conductive member are integral components of a same wire.

(15) In some embodiments, the conductive coil and the elongate conductive member are parts of different wires.

(16) In some embodiments, the conductive tubular element includes a conductive coil and a conductive cannula disposed over the conductive coil.

(17) In some embodiments, the elongate conductive member longitudinally extends outside and along an outer surface of the body from the proximal portion to the distal portion.

(18) In some embodiments, the elongate conductive member longitudinally extends inside the body from the proximal portion to the distal portion. The distal portion of elongate conductive member extends from inside to outside of the elongate tubular member to be attached to the conductive tubular member.

(19) In some embodiments, the elongate conductive member is embedded in the body.

(20) In some embodiments, the elongate conductive member is in a straightened configuration while embedded in the body.

(21) In some embodiments, the elongate conductive member is in a coiled configuration while embedded in the body.

(22) In some embodiments, the wireguide lumen includes a first lumen, and the elongate tubular member further includes a second lumen longitudinally extending in the body from the proximal portion to the distal portion. The elongate tubular member longitudinally extends in the second lumen from the proximal portion to the distal portion.

(23) In some embodiments, an outer diameter of the conductive tubular element is greater than five French (one and two-thirds millimeters) and less than ten French (three and one-third millimeters).

(24) In some embodiments, the outer diameter of the conductive tubular element is six French.

(25) In another embodiment, a method of gaining access into an anatomical structure may be

performed. The method may include delivering a conductive tubular element to a treatment site in a patient; electrically activating the conductive tubular element; and creating an opening in the anatomical structure by initially cutting the anatomical structure with the electrically activated conductive tubular element.

(26) In some embodiments, creating the opening includes creating an incision in the anatomical structure.

(27) In some embodiments, creating the opening includes enlarging an initial opening of the anatomical structure.

(28) In some embodiments, creating the opening includes creating the opening with only the electrically activated conductive tubular element.

(29) In some embodiments, initially cutting into the anatomical structure comprises distally advancing a distal end of the conductive tubular element into the anatomical structure while the conductive tubular element is electrically activated.

(30) In some embodiments, initially cutting into the anatomical structure includes initially cutting into the anatomical structure with a side surface defining a distal end of the conductive tubular element.

(31) In some embodiments, the method further includes: after creating the opening, positioning a drainage device in the opening for drainage of the anatomical structure.

(32) In some embodiments, the method further includes: before positioning the drainage device in the opening, distally advancing a distal portion of a wireguide through the opening to within the anatomical structure; and distally advancing the drainage device over the wireguide to the treatment site in order to position the drain device in the opening.

(33) In some embodiments, the conductive tubular element is disposed about an elongate tubular member, distally advancing the distal portion of the wireguide comprises distally advancing the distal portion through a wireguide lumen of the elongate tubular member.

(34) In some embodiments, an outer diameter of the conductive tubular element is greater than five French (one and two-thirds millimeters) and less than ten French (three and one-third millimeters).

(35) In some embodiments, the conductive tubular element includes at least one of a conductive cannula or a conductive coil.

(36) In some embodiments, the method further includes: delivering electrical current in a range of 80-120 Watts to the conductive tubular element while the conductive tubular element is cutting into the anatomical structure to create the opening.

(37) In some embodiments, delivering the conductive tubular element to the treatment site comprises delivering the conductive tubular element disposed over an elongate tubular member comprising a body and a wireguide lumen longitudinally extending through the body, where the body includes an insulating material that electrically insulates the wireguide lumen from the conductive tubular element.

(38) In some embodiments, the treatment site is in a gastrointestinal tract of the patient.

(39) In some embodiments, the anatomical structure is adjacent a wall of the gastrointestinal tract, and the method further includes cutting into the wall before cutting into the anatomical structure in order to create the opening.

(40) In some embodiments, the anatomical structure includes a pseudocyst.

(41) In some embodiments, the pseudocyst is adjacent a wall outside the gastrointestinal tract.

(42) Other embodiments are possible, and each of the embodiments can be used alone or together in combination. Accordingly, various embodiments are described below with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a side view of a distal portion of a prior art cystotome delivered to a treatment site near a pseudocyst.
- (2) FIG. 1B is a side view of the distal portion of the prior art cystotome of FIG. 1A, with a needle knife disposed in an inner elongate tubular member having cut into the pseudocyst.
- (3) FIG. 1C is a side view of the distal portion of the prior art cystotome of FIG. 1A, with a conductive cannula coupled to an outer elongate tubular member having further cut into the pseudocyst and a distal portion of a wireguide being inserted into the cystotome.
- (4) FIG. 1D is a side view of the distal portion of the wireguide being left in the pseudocyst with the distal portion prior art cystotome of FIG. 1A having been removed from the treatment site.
- (5) FIG. 2 is a partial cross-sectional side view of a first example electrosurgical device configured to gain access into a pseudocyst that includes a conductive cannula and an elongate conductive member longitudinally extending on an outer surface of an elongate tubular member.
- (6) FIG. 3 is a partial cross-sectional side view of a second example electrosurgical device configured to gain access into a pseudocyst that includes a conductive cannula and an elongate conductive member longitudinally extending while embedded in a body of an elongate tubular member in a generally straightened manner.
- (7) FIG. 4 is a partial cross-sectional side view of a third example electrosurgical device configured to gain access into a pseudocyst that includes a conductive cannula and an elongate conductive member embedded in a body of an elongate tubular member in a coiled configuration.
- (8) FIG. 5 is a partial cross-sectional side view of a fourth example electrosurgical device configured to gain access into a pseudocyst that includes a conductive cannula and an elongate conductive member longitudinally extending in a lumen of the elongate tubular member.
- (9) FIG. 6 is a partial cross-sectional side view of a fifth example electrosurgical device configured to gain access into a pseudocyst that includes a conductive coil integral with an elongate tubular member longitudinally extending on an outer surface of an elongate tubular member.
- (10) FIG. 7 is a partial cross-sectional side view of a sixth example electrosurgical device configured to gain access into a pseudocyst that includes a conductive coil integral with an elongate tubular member that longitudinally extends in a generally straightened manner while embedded in a body of an elongate tubular member.
- (11) FIG. 8 is a partial cross-sectional side view of a seventh example electrosurgical device configured to gain access into a pseudocyst that includes a first conductive coil integral with a second conductive coil embedded in a body of an elongate tubular member.
- (12) FIG. 9 is a partial cross-sectional side view of an eight example electrosurgical device configured to gain access into a pseudocyst that includes a conductive coil integral with an elongate conductive member longitudinally extending in a lumen of an elongate tubular member.
- (13) FIG. 10 is a partial cross-sectional side view of a ninth example electrosurgical device configured to gain access into a pseudocyst that includes a conductive coil disposed over a distal end of a separate elongate conductive member longitudinally extending on an outer surface of an elongate tubular member.
- (14) FIG. 11 is a partial cross-sectional side view of a tenth example electrosurgical device configured to gain access into a pseudocyst that includes a conductive coil disposed over a distal end of a separate elongate conductive member longitudinally extending in a generally straightened manner while embedded in a body of an elongate tubular member.
- (15) FIG. 12 is a partial cross-sectional side view of an eleventh example electrosurgical device configured to gain access into a pseudocyst that includes a conductive coil disposed over a distal end of a separate elongate conductive member in a coiled configuration while embedded in a body of an elongate tubular member.
- (16) FIG. 13 is a partial cross-sectional side view of a twelfth example electrosurgical device

configured to gain access into a pseudocyst that includes a conductive coil disposed over a distal end of a separate elongate conductive member longitudinally extending in a lumen of an elongate tubular member.

(17) FIG. 14 is a partial cross-sectional side view of a thirteenth example electrosurgical device configured to gain access into a pseudocyst that includes a conductive coil integral with an elongate conductive member longitudinally extending on an outer surface of an elongate tubular member, and a conductive cannula disposed over the conductive coil.

(18) FIG. 15A is a side view of the first example electrosurgical device of FIG. 2, with its distal portion being delivered to a treatment site near a pseudocyst.

(19) FIG. 15B is a side view of the first example electrosurgical device of FIG. 2, with its distal portion creating an incision in a wall and the pseudocyst of FIG. 15A.

(20) FIG. 15C is a side view of the first example electrosurgical device of FIG. 2, with a distal portion of a wireguide being inserted into the pseudocyst of FIG. 15A.

(21) FIG. 15D is a side view of the distal portion of the wireguide of FIG. 15C being kept inside the pseudocyst of FIG. 15A, while a distal portion of the first example electrosurgical device of FIG. 2 has been withdrawn from the treatment site.

(22) FIG. 15E is a partial cross-sectional side view of a drainage device being positioned in the created incision.

DETAILED DESCRIPTION

(23) The present disclosure describes various exemplary embodiments of an electrosurgical device that includes a conductive distal tip that is configured to create an opening in order to gain access into an anatomical structure, such as one that is in or adjacent to the gastrointestinal tract of a patient. The present disclosure also describes related methods of gaining access into anatomical structure using the various exemplary embodiments of the electrosurgical device.

(24) The conductive distal tip may include a conductive tubular or cylindrical element disposed about an outer surface of an elongate tubular member, such as a catheter, at a distal portion of the elongate tubular member. The cylindrical conductive distal tip may be made of a conductive material, such as stainless steel or copper as non-limiting examples. Other conductive materials may be possible. In some example embodiments, the conductive tubular element may be a cannula that is a continuous, solid structure and/or that has a generally smooth outer surface over its longitudinal length. In other example embodiments, the conductive tubular element may be a coil or coiled structure, such as a coiled wire, that is a discontinuous structure over its longitudinal length and/or that has a generally ribbed outer surface over its longitudinal length as formed by the coils. In still other example embodiments, the conductive tubular element may be combination of the cannula and the coil. A longitudinal length of the conductive tubular element may be in a range from three to eight millimeters, although other longitudinal lengths may be possible.

(25) In addition to being disposed about the elongate tubular member, the conductive distal tip may also be affixed to the elongate tubular member. By being affixed to the elongate tubular member, the distal tip may longitudinally move with the elongate tubular member. In other words, the elongate tubular member may be longitudinally moved in order to longitudinally move the conductive distal tip.

(26) The tubular conductive distal tip may have an outer diameter, and/or the tubular conductive distal tip in combination with the elongate tubular member, may have a total outer diameter that is greater than five French (1.67 millimeters) and less than 10 French (3.33 millimeters). In one example, the total outer diameter is 6 French. In some example embodiments, the elongate tubular member itself may have an outer diameter of five French, and the total outer diameter of the conductive distal tip and the elongate tubular member may be five French plus the thickness of the conductive distal tip. An example thickness of the conductive distal tip may be four one-thousandths of an inch (0.1016 millimeters), although other thicknesses may be possible. With an outer diameter in between five French and ten French, the tubular conductive distal tip may be

suitably sized to make an optimally sized incision into a pseudocyst for insertion of a stent or other drainage device having a size or outer diameter of less than ten French, without the need to additionally use other-sized conductive elements to assist in making the incision.

(27) The conductive distal tip may be attached and electrically connected to an elongate conductive member, such as a wire, that longitudinally extends with the elongate tubular member from a proximal portion to the distal portion. Like the conductive distal tip, the elongate conductive member may be made of a conductive material, such as stainless steel or copper as non-limiting examples. The conductive distal tip may be fixedly attached to the elongate conductive member through one or more attachment mechanism other than solder. The attachment mechanism(s) used may depend on the configuration of the conductive distal tip. Where the conductive distal tip includes a cannula, the cannula may be crimped to the elongate conductive member, welded to the elongate conductive member, or a combination thereof. Where the conductive distal tip includes a coil, the coil and the elongate conductive member may be part of and/or formed from different wires, in which case the coil may be crimped and/or welded to the elongate conductive member, similar to how the cannula is attached to the elongate conductive member. Alternatively, the coil and the elongate conductive member may be part of and/or formed from the same wire, in which case the coil and the elongate conductive member are attached to each other by virtue of being part of and/or formed from the same wire.

(28) The elongate conductive member may longitudinally extend with and/or adjacent to the body of the elongate tubular member in various ways. In some example embodiments, the elongate conductive member may extend along the outer surface and/or outside of the elongate tubular member. In other example embodiments, the elongate conductive member may longitudinally extend within the elongate tubular member. When extending within the elongate tubular member, a distal portion of the elongate conductive member may extend to outside of the elongate tubular member, where it may be attached and electrically connected to the conductive distal tip.

Additionally, for some embodiments where the elongate conductive member extends within the elongate tubular member, the elongate conductive member may be embedded in a body of the elongate tubular member. For other embodiments, the elongate conductive member may be disposed and longitudinally extend in a lumen of the elongate tubular member. Where the elongate conductive member is embedded in the body, the conductive member may longitudinally extend in the body in a generally straightened configuration or in a coiled configuration.

(29) FIGS. 2-14 show various example embodiments of an electrosurgical device that includes a conductive tubular distal tip attached and electrically connected to an elongate conductive member in accordance with the description provided above. The electrosurgical device may be part of an electrosurgical medical system along with a power source configured to supply electrical current to the distal tip.

(30) Referring to FIG. 2, a partial cross-sectional side view of a first example embodiment of an electrosurgical device **202** in electrical connection with a power source **204** is shown. The power source **204** may be an electronic device, such as a radio frequency (RF) generator or an electrosurgical unit (ESU) that is configured to generate and supply electrical current, including RF electrical current, to the electrosurgical device **202**.

(31) The electrosurgical device **202** may include an elongate tubular member **206**, such as a catheter, that longitudinally extends from a proximal portion **208** to a distal portion **210**. The elongate tubular member **206** may include a body **212** and a wireguide lumen **214** longitudinally extending in the body **212** from the proximal portion **208** to the distal portion **210**. The wireguide lumen **214** may be sized to have a wireguide movably disposed therein. An example size (i.e., diameter) of the wireguide lumen **214** may be in a range of about 0.010 inches to 0.040 inches. Particular example sizes may include 0.035 inches or 0.038 inches. The body **212** may be made of an insulating material such as polyether ether ketone (PEEK) or polytetrafluoroethylene (PTFE) as examples, although other insulating materials may be possible.

(32) The electrosurgical device **202** may further include a conductive cannula **216** disposed on or about an outer surface **218** of the elongate tubular member **206** at the distal portion **210**. In one example embodiment, as shown in FIG. 2, a distal end **217** of the conductive cannula **216** may be flush or even with a distal end **219** of the elongate tubular member **206**. This way, when the distal portion is delivered to a treatment site, a side surface of cannula **216** defining the distal end **217** may come into contact with an anatomical structure, so that when activated by electrical energy, the side surface is able to cut the tissue of the anatomical structure, as described in further detail below. In other example embodiments, the distal end **217** of the cannula **216** may extend distally past the distal end **219** of the elongate tubular member **206**. The insulating material of the body **212** of the tubular member **206** may insulate the wireguide lumen **214** (or a wireguide inside the wireguide lumen) from the conductive cannula **216**.

(33) The conductive cannula **216** may be configured to be electrically activated by the power source **204** via an elongate conductive member **220**, such as a wire. The elongate conductive member **220** may longitudinally extend along or adjacent to the outer surface **218** from the proximal portion **208** to the distal portion **210**. At the distal portion **210**, a distal end **222** of the elongate conductive member **220** may be fixedly attached and electrically connected to the conductive cannula **216**. As shown in FIG. 2, the distal end **222** may lay or otherwise be disposed on the outer surface **218**, and the conductive cannula **216** may be disposed over and/or cover the distal end **222**. The conductive cannula **216** may be then fixedly attached to the distal end **222** by performing a mechanical process (e.g., a crimping process that crimps the conductive cannula **216** to the distal end **222**) and/or a thermal process (e.g., a welding process that welds the conductive cannula **216** to the distal end **222**). The conductive cannula **216** may be fixedly attached to the distal end **222** without the use of solder, although in some configurations, solder may be added to enhance the fixed attachment.

(34) The electrosurgical device **202** may further include a handle assembly **224** operatively coupled to proximal ends **226**, **228** of the elongate tubular member **206** and the elongate conductive member **220**. The elongate conductive member **220** may be electrically coupled to the power source **204** via the handle assembly **224**. For example, as shown in FIG. 2, the handle assembly **224** may include a conductive coupling element **230** configured to be electrically coupled to the proximal end **228** of the elongate conductive member **220**. The handle assembly **224** may also include a connector **232** adapted to be connected to electrical cabling **234**, which may also be connected to an output port **236** of the power source **204**. For some example configurations, the electrical cabling **234** may be adapted to be removably connected to the handle connector **232** and/or the output port **236**. When the electrical cabling **234** is connected to the output port **236** of the power source **204** and the connector **232** of the handle assembly **224**, the power source **204** may be configured to deliver electrical current to the elongate conductive member **220** via the electrical cabling **234** and the conductive coupling element **234** of the handle assembly **224**. In turn, the elongate conductive member **220** may deliver the electrical current to the conductive cannula **216**. In addition, as shown in FIG. 2, the handle assembly **224** may include a wireguide portion **250** in communication with the wireguide lumen **214** for insertion of a wireguide into the wireguide lumen **214** and removal therefrom.

(35) For some example configurations, the elongate tubular member **206** may be an inner elongate tubular member **206**, and the electrosurgical device **202** may further include an outer elongate tubular member **238** coaxial with the inner elongate tubular member **206**. The outer elongate tubular member **238** may longitudinally extend from the proximal portion **208** to the distal portion **210**, and a proximal end **240** may be coupled to the handle assembly **224**. The outer elongate tubular member **238** may include a body **242** and a central lumen **244** longitudinally extending in the body **242**. The inner elongate tubular member **206** may longitudinally extend within the central lumen **244** from the proximal portion **208** to the distal portion **210**. The elongate conductive member **220** may longitudinally extend in between the outer surface **218** of the inner elongate

tubular member **206** and an inner surface **246** of the body **240** of the outer elongate tubular member **238** defining the central lumen **242**.

(36) The handle assembly **224** may be configured to longitudinally move the inner elongate tubular member **206**, the conductive cannula **216**, and the elongate conductive member **220** relative to the outer elongate tubular member **238**. During operation, when the distal portion **206** is being delivered to a treatment site within a patient, the handle assembly **224** may move the inner and outer elongate tubular members **206**, **238** relative to each other so that the conductive cannula **216** is disposed inside the outer member **238**, such as within the central lumen **244**. Subsequently, at the treatment site, the handle assembly **224** may move the inner and outer elongate tubular members **206**, **238** relative to each other so that the conductive cannula **216** is distally past a distal end **248** of the outer member **238** and exposed to its outer surroundings so that the conductive cannula **216** may contact tissue to perform the electrosurgical procedure. For other example configurations, the electrosurgical device **202** may not include the outer elongate tubular member **238**.

(37) Referring to FIG. 3, a partial cross-sectional side view of a second example embodiment of an electrosurgical device **302** in electrical connection with the power source **204** is shown. The electrosurgical device **302** of FIG. 3 may be similar to the electrosurgical device **202** of FIG. 2, except that, as shown in FIG. 3, the electrosurgical device **302** may include an elongate conductive member **320** that is embedded in a body **312** of an elongate tubular member **306**. While embedded in the body **312**, the elongate conductive member **320** may longitudinally extend from a proximal portion **308** to a distal portion **310** in a generally straightened manner. At the distal portion **310**, the elongate conductive member **320** may extend to outside the elongate tubular member **306**, and a distal portion **322** may lay or otherwise be disposed on an outer surface **318** of the elongate tubular member **306**. Like the electrosurgical device **202** of FIG. 2, a conductive cannula **316** may be disposed about the outer surface **318** of the elongate tubular member **306** and over the distal end **322** of the elongate conductive member **306**. The conductive cannula **316** may be fixedly attached to the distal end **322**, such as by use of a mechanical (e.g., crimping) and/or a thermal (e.g., welding) process.

(38) Referring to FIG. 4, a partial cross-sectional side view of a third example embodiment of an electrosurgical device **402** in electrical connection with the power source **204** is shown. The third embodiment **402** is similar to the second and third embodiments **202**, **302**, except that, as shown in FIG. 4, the electrosurgical device **402** may include an elongate conductive member **420** that is in a coiled configured while embedded and longitudinally extending in a body **412** of an elongate tubular member **406** from a proximal portion **408** to a distal portion **410**. A distal portion **422** of the embedded coil **420** may extend from the body **412** to an outer surface **418** of the elongate tubular member **406**, where a conductive cannula **416** may be disposed over and fixedly attached to the distal portion **422**. FIG. 4 shows the distal portion **422** in a generally straightened configuration when disposed outside the elongate tubular member **406**, although in other configurations, the distal portion **422** may maintain the coiled configuration when disposed outside and on the outer surface **418**.

(39) Referring to FIG. 5, a partial cross-sectional side view of a fourth example embodiment of an electrosurgical device **502** in electrical connection with the power source **204** is shown. The fifth example embodiment **502** may be similar to the second example embodiment **302** of FIG. 3, except instead of being embedded in the body, the fourth example embodiment may include a lumen in which the elongate conductive member may be disposed. In particular, the fifth example embodiment of the electrosurgical device **502** may include two lumens longitudinally extending in a body **512** of an elongate tubular member **506** from a proximal portion **508** to a distal portion **510**. The two lumens may include a wireguide lumen **514** and a second lumen **540** for an elongate conductive member **520**. The elongate conductive member **520** may longitudinally extend in the second lumen **540** from the proximal portion **508** to the distal portion **510**. At the distal portion **510**, a distal portion **522** of the elongate conductive member **520** may exit the second lumen **540** and distally extend to outside the elongate tubular member **506**, where the distal portion **522** may

be disposed on an outer surface **518** of the elongate tubular member **506**. Like the first through third example embodiments, a conductive cannula **516** may be disposed over and fixedly attached to the distal portion **522**.

(40) Referring to FIG. **6**, a partial cross-sectional side view of a fifth example embodiment of an electrosurgical device **602** in electrical connection with the power source **204** is shown. Rather than include a conductive cannula as in the first through fourth example embodiments of FIGS. **2-5**, the fifth example embodiment of the electrosurgical device **602** may include a conductive coil **616** disposed on or about an outer surface **618** of an elongate tubular member **606** at a distal portion **610**. Like the first example embodiment **202** of FIG. **2**, the electrosurgical device **602** may also include an elongate conductive member **620** that longitudinally extends along the outer surface **618** from a proximal portion **608** to the distal portion **610**. A proximal end **628** of the elongate conductive member **620** may be electrically coupled to the coupling element **230** of the handle assembly **224** in order to electrically couple the conductive coil **616** with the power source **204**. In the fifth example embodiment **602** of FIG. **6**, the conductive coil **616** and the elongate conductive member **620** may be integral components of the same wire. Also, FIG. **6** shows the elongate conductive member **620** extending from the proximal portion **608** to the distal portion **610** in a generally straightened configuration. In other example configurations, the elongate conductive member **620** may have coiled configuration as it extends from the proximal portion **608** to the distal portion **610**.

(41) Referring to FIG. **7**, a partial cross-sectional side view of a sixth example embodiment of an electrosurgical device **702** in electrical connection with the power source **204** is shown. The sixth example embodiment **702** may be similar to the fifth example embodiment **602** of FIG. **6**, except that the electrosurgical device **702** may include an elongate conductive member **720** embedded in a body **712** of an elongate tubular member **706** while longitudinally extending in the body **712** from a proximal portion **708** to a distal portion **710**. As shown in FIG. **7**, the elongate conductive member **720** may longitudinally extend in the body **712** in a generally straightened manner. In addition, like the fifth example embodiment **602** of FIG. **6**, the elongate conductive member **720** and a conductive coil **716** may be integral components of the same wire.

(42) Referring to FIG. **8**, a partial cross-sectional side view of a seventh example embodiment of an electrosurgical device **802** in electrical connection with the power source **204** is shown. The electrosurgical device **802** may include a first conductive coil **816** disposed on an outer surface **818** of an elongate tubular member **806** at a distal portion **810**. The electrosurgical device **802** may also include a second conductive coil **820** embedded in a body **812** of the elongate tubular member **806**. The second conductive coil **820** may extend from a proximal portion **808** to the distal portion **810**. In the example embodiment **802**, the first conductive coil **816** and the second conductive coil **820** may be integral components of the same wire. At the distal portion **810**, the wire may extend from the body **812** to outside the elongate tubular member **806**, where the wire forms the first conductive coil disposed on the outer surface **818**.

(43) Referring to FIG. **9**, a partial cross-sectional side view of an eighth example embodiment of an electrosurgical device **902** in electrical connection with the power source **204** is shown. The eighth example embodiment **902** may include a wireguide lumen **914** and a second lumen **940** longitudinally extending in a body **912** from a proximal portion **908** to a distal portion **910** of an elongate tubular member **906**. An elongate conductive member **920** may be disposed and longitudinally extend within the second lumen **920** from the proximal portion **908** to the distal portion **910**. Similar to the fifth, sixth, and seventh example embodiments **602**, **702**, **802** of FIGS. **6-8**, the elongate conductive member **920** and a conductive coil **916** may be integral components of the same wire. At the distal portion **910**, the elongate conductive member **920** may exit the second lumen **940** and extend to outside the elongate tubular member **906** to form the conductive coil **916** disposed on an outer surface **918** of the elongate tubular member **906**.

(44) FIGS. **10-13** are partial cross-sectional side views of ninth, tenth, eleventh, and twelfth

example embodiments of electrosurgical devices **1002**, **1102**, **1202**, **1303**, respectively. These example embodiments are similar to the first, second, third, and fourth example embodiments **202**, **302**, **402**, **502**, respectively, except that they each include a conductive coil as the conductive distal tip instead of a conductive cannula. In each of the embodiments, though, the conductive coil and a respective elongate conductive member may be parts of different wires. Accordingly, the conductive coils may be attached to their respective elongate conductive members through a mechanical process (e.g., crimping) and/or a thermal process (e.g., welding).

(45) In further detail, referring to FIG. **10**, an elongate conductive member **1020** may longitudinally extend along an outer surface **1018** of an elongate tubular member **1006** from a proximal portion **1008** to a distal portion **1010**. At the distal portion **1010**, a conductive coil **1016** may cover and be fixedly attached and electrically connected to a distal portion **1022** of the elongate conductive member **1020**.

(46) Referring to FIG. **11**, an elongate conductive member **1120** may be embedded and longitudinally extend in a body **1112** of an elongate tubular member **1106** from a proximal portion **1108** to a distal portion **1110**. At the distal portion **1110**, a distal portion **1122** of the elongate conductive member **1120** may extend from the body **1112** to outside the elongate tubular member **1106**, where the distal portion **1122** is disposed on an outer surface **1118**. A conductive coil **1116** may cover and be fixedly attached and electrically connected to the distal portion **1122** of the elongate conductive member **1120**.

(47) Referring to FIG. **12**, a first conductive coil **1220** may be embedded in a body **1212** of an elongate tubular member **1206**, and while embedded, longitudinally extend from a proximal portion **1208** to a distal portion **1210**. At the distal portion **1210**, a distal portion **1222** of the first conductive coil **1220** may extend from the body **1212** to outside the elongate tubular member **1206**, where the distal portion **1222** may be disposed on an outer surface **1228**. A second conductive coil **1216** may cover and be fixedly attached and electrically connected to the distal portion **1222** of the first conductive coil **1220**.

(48) Referring to FIG. **13**, an elongate tubular member **1306** may include a wireguide lumen **1314** and a second lumen **1340** longitudinally extending in a body **1312** from a proximal portion **1308** to a distal portion **1310**. An elongate conductive member **1320** may be disposed and longitudinally extend in the second lumen **1340** from the proximal portion **1308** to the distal portion **1310**. At the distal portion **1310**, a distal portion **1322** of the elongate conductive member **1320** may exit the second lumen **1340** and extend to outside the elongate tubular member **1306**, where the distal portion **1322** may be disposed on an outer surface **1318** of the elongate tubular member **1306**. A conductive coil **1316** may cover and be fixedly attached and electrically connected to the distal portion **1322** of the elongate conductive member **1320**.

(49) Referring to FIG. **14**, a partial cross-sectional side view of a thirteenth example embodiment of an electrosurgical device **1402** in electrical connection with the power source **204** is shown. The electrosurgical device **1402** may be similar to the fifth example embodiment **602** of FIG. **6**, except that the thirteenth example embodiment **1402** may further include a conductive cannula **1442** disposed over and/or about the conductive coil **1416**. The conductive cannula **1442** may be fixedly attached and electrically connected to the conductive coil **1416** through a mechanical mechanism (e.g., crimping) and/or a thermal mechanism (e.g., welding).

(50) The thirteen embodiments shown in FIGS. **2-14** are exemplary and other configurations including various combinations of the above-described features may be possible. For example, the sixth, seventh, and eighth example embodiments **702**, **802**, **902** of FIGS. **7**, **8**, and **9**, respectively, may similarly include a conductive cannula disposed over a conductive coil to form a conductive tubular distal tip, similar to the thirteenth example embodiment shown in FIG. **14**. Other configurations may be possible. As another example, each of the embodiments in FIGS. **2-14** are shown as including the outer elongate tubular member **238**. However, alternative configurations for any of these embodiments may not include the outer elongate tubular member **238**. For these

alternative configurations, the handle assembly **224** may still be operative to maneuver the single tubular member and the conductive tubular distal tip and to deliver electrical current to the conductive distal tip, but may not necessarily be adapted to move coaxial inner and outer tubular members relative to each other.

(51) A method of gaining access by creating an opening into an anatomical structure that is in, adjacent to, and/or accessible from a treatment site of a patient is described with reference to FIGS. **15A-15E**. The method described with reference to these figures identifies a treatment site **1500** as being an area of a gastrointestinal (GI) tract, and an anatomical structure **1502** as being a pseudocyst that is outside and adjacent to the GI tract and pushing up against a wall **1504** of the GI tract. However, the method may be similarly applied to gaining access into anatomical structures other than pseudocysts and/or anatomical structure that are located in other areas of or within a body of a patient, including those that are in, adjacent to, and/or accessible from the GI tract. An example anatomical structure may be the papilla of Vater, which is an exit hole for the biliary tree that delivers secretions from organs such as the liver, gallbladder, and pancreas, into the duodenum. Other anatomical structures may be possible. For some anatomical structures, such as a pseudocyst, creating the opening may include making or creating an incision in the anatomical structure. For other anatomical structures, such as the papilla of Vater, the anatomical structure may already have an initial opening. However, the initial opening is too small and/or restrictive for access into an area beyond the anatomical structure, such as access into the biliary tree from the duodenum. For these other anatomical structures, creating the opening includes enlarging the initial or pre-existing opening by performing the cutting. After the opening is created, a subsequent act may be performed that is related to the purpose of gaining access into the anatomical structure. For example, with pseudocysts, a drainage device, such as a stent, may be inserted into the opening in order to drain the stent. As another example, a medical device (e.g., forceps or a basket) maybe inserted into the opening in order to remove stones (e.g., gallstones), or a visualization device (e.g., a camera or image sensor) may be inserted through the opening for visual diagnostic purposes. Other actions after the opening is created may be performed. Also, the method is described as being performed using the first embodiment **202** of the electrosurgical device shown and described with reference to FIG. **2**, although any of the other embodiments may be similarly used to perform the method.

(52) Referring to FIG. **15A**, the distal portion **210** of the elongate tubular member **206** may be delivered to a treatment site **1500** within a patient, such as an area of the patient's GI tract. Although not shown, the distal portion **210** may be delivered to the treatment site **1500** by being moved through a channel or lumen of an endoscope. At the treatment site **1500**, a pseudocyst **1502**, such as a pancreatic pseudocyst, may be pushing up against a wall **1504** (such as a transgastric or transduodenal wall) of an internal organ associated with the GI tract, such as the stomach or the duodenum. In some methods, the pseudocyst **1502** may be identified by identifying a bulge in the wall **1504** using a visualization device (not shown). In some methods, while being delivered to the treatment site **1500**, the conductive cannula **216** may be disposed within the outer elongate tubular member **238**. At the treatment site **1500**, the handle assembly **224** may be operated to move the conductive cannula **216** distally past the distal end **248** of the outer elongate tubular member **238** so that the conductive cannula **216** is exposed to the tissue at the treatment site **1500**, which is shown in FIG. **15A**. Alternatively, where an electrosurgical device not including the outer elongate tubular member **238** is used to perform the method, this operation of the handle assembly **224** to expose the conductive cannula **216** may not be performed.

(53) In addition, when the conductive cannula **216** is at the treatment site **1500**, the conductive cannula **216** may be electrically activated by activating the power source **204**. When activated, the power source **204** may deliver electrical current from the output port **236**, through the cabling **234**, through the conductive coupling component **230**, through the elongate conductive member **220**, and to the conductive cannula **216**. Power associated with the electrical current that is delivered may be in a range of between 80-120 Watts, although other power settings may be possible. Also,

the frequency of the electrical current may be a frequency used for electrosurgery, which may be in the radio frequency (RF) range. An example frequency range may include 350 kiloHertz (kHz) to 4 Megahertz (MHz), although other frequencies used for electrosurgery outside of this range may be possible.

(54) Referring to FIG. 15B, the elongate tubular member **206** and the electrically activated conductive cannula **216** may be distally advanced so that the conductive cannula **216**, including its distal end **217**, contacts and cuts into the bulging part of the wall **1504**. The tubular member **206** and the conductive cannula **216** may continue to be distally advanced so that the conductive cannula **216** further cuts into the pseudocyst **1502** to make an opening, such as an incision, **1508**. (As previously described, other anatomical structures may already include an opening **1508** before the cutting, such that creating the opening may include enlarging the existing opening **1508**). As shown in FIG. 15B, at this point during the procedure, a distal portion of the tubular member **206** and conductive cannula **216** may have gained access into and be disposed in the pseudocyst **1502**. For other methods involving treatments sites other than the one shown in FIGS. 15A-15E, where there is no wall **104** separating the anatomical structure from the conductive cannula **216** (e.g., where the anatomical structure is in the treatment site (such as on or attached to the wall **104** and protruding into the GI tract), the cutting operation may not involve cutting into the wall **104**, and instead may involve cutting directly into the anatomical structure.

(55) Referring to FIG. 15C, with the tubular member **206** and conductive cannula **216** having gained access into and disposed in the pseudocyst **1504**, a wireguide **1510** may be distally advanced through the wireguide lumen **214** until a sufficient amount of a distal portion **1512** of the wireguide **1510** is distally advanced past distal ends **252**, **254** of the tubular member **206** and the cannula **216** inside the pseudocyst **1502**. FIG. 15C shows the distal portion **1512** having curled inside the pseudocyst **1502** when past the distal end **252** and outside of the tubular member **206**. As shown in FIG. 15C, the wireguide **1510** may be inserted into the wireguide portion **250** of the handle assembly **224** and then advanced into the wireguide lumen **214** from the handle assembly **224**. Also, in some example methods, the conductive cannula **216** may be electrically deactivated from the power source **204** prior to the distal portion **1512** being advanced past the distal ends **252**, **254** of the tubular member **206** and the conductive cannula **216** in order to ensure that the distal portion **1512** is not electrically activated. During the operation, if the physician determines that further cutting with the conductive cannula **216** is to be performed, the distal portion **1512** may be withdrawn back to within the wireguide lumen **214**, where the body **212** may electrically insulate the distal portion **1512** from the electrically activated conductive cannula **216**. Alternatively, the distal portion **1512** and the conductive cannula **216** may be moved relative to each other within the pseudocyst **1502** to be a sufficient distance away from each other so that when the cannula **216** is electrically activated for further cutting, the wireguide **1510** is not also electrically activated. Electrically activating the wireguide **1510** may be undesirable as it could possible harm to the patient and/or the physician maneuvering the wireguide **1510**.

(56) Referring to FIG. 15D, after the distal portion **1512** of the wireguide **1510** is positioned in the pseudocyst **1502** and cutting with the conductive cannula **216** is complete, the conductive cannula **216** and the distal portion **210** of the tubular member **206** may be withdrawn from inside the pseudocyst **1502** and further to outside the patient, while wireguide **1510** remains in the patient, with the distal portion **1512** disposed in the pseudocyst **1502**. Referring to FIG. 15E, a stent or other drainage device **1514** may be inserted onto the wireguide **1510** outside of the patient and distally advanced within the patient to the treatment site **1500**. The drainage device **1514** may be further advanced and positioned in the opening **1508**, where at least a portion of the drainage device **1514** is disposed within the pseudocyst **1502**.

(57) The drainage device **1504** may have an outer diameter of less than 10 French. As described above, the conductive cannula **216** may have an outer diameter in between 5 French and 10 French, such as 6 French, which may provide a suitable size of the opening **1508** for the drainage device

1514 to be securely held and/or maintained inside the pseudocyst **1502**. Also, in the example method described with reference to FIGS. **15A-15E**, only a single conductive tubular element affixed to a single tubular member is electrically activated to create the opening **1508**. That is, a second conductive tubular element affixed to a second tubular member is not used to make the opening **1508**. With reference to FIGS. **15A-15E**, the single conductive tubular element is the cannula **216**, although in other methods, the single conductive tubular element may be a coil or a combination of a cannula and a coil, as shown in described with reference to FIGS. **3-14**.

(58) After the drain device **1514** is inserted into the opening **1508**, the wireguide **1510** may be removed from the treatment area **1500** and drainage of the pseudocyst **1502** may be performed via the drainage device **1514**.

(59) The foregoing description of various embodiments of the invention has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise embodiments disclosed. Numerous modifications or variations are possible in light of the above teachings. The embodiments discussed were chosen and described to provide the best illustration of the principles of the invention and its practical application to thereby enable one of ordinary skill in the art to utilize the invention in various embodiments and with various modifications as are suited to the particular use contemplated. All such modifications and variations are within the scope of the invention as determined by the appended claims when interpreted in accordance with the breadth to which they are fairly, legally, and equitably entitled.

Claims

1. A method of gaining access into a pseudocyst, the method comprising: delivering an electrosurgical device separate from an endoscope to a treatment site, the electrosurgical device delivered to the treatment site via a channel or lumen of the endoscope, the electrosurgical device comprising an outer elongate tubular member, an inner elongate tubular member, and a conductive tubular element disposed on an outer surface of the inner elongate tubular member, wherein an outer diameter of the conductive tubular element is greater than five French (one and two-thirds millimeters) and less than ten French (three and one-third millimeters); electrically activating the conductive tubular element; and creating an opening in the pseudocyst by initially cutting the pseudocyst with the electrically activated conductive tubular element and moving the electrically activated conductive tubular element into the pseudocyst while leaving the outer elongate tubular member outside of the pseudocyst, wherein the pseudocyst does not have any opening from cutting prior to the initial cutting with the electrically activated conductive tubular element.
2. The method of claim 1, wherein creating the opening comprises creating an incision in the pseudocyst.
3. The method of claim 1, wherein creating the opening comprises creating the opening with only the electrically activated conductive tubular element.
4. The method of claim 1, wherein initially cutting into the pseudocyst comprises distally advancing a distal end of the conductive tubular element into the pseudocyst while the conductive tubular element is electrically activated.
5. The method of claim 1, wherein initially cutting into the pseudocyst comprises initially cutting into the pseudocyst with a side surface defining a distal end of the conductive tubular element.
6. The method of claim 1, further comprising: after creating the opening, positioning a drainage device in the opening for drainage of the pseudocyst.
7. The method of claim 6, further comprising: before positioning the drainage device in the opening, distally advancing a distal portion of a wireguide through the opening to within the pseudocyst; and distally advancing the drainage device over the wireguide to the treatment site in order to position the drain device in the opening.
8. The method of claim 7, wherein distally advancing the distal portion of the wireguide comprises

distally advancing the distal portion through a wireguide lumen of the elongate tubular member.

9. The method of claim 1, wherein the conductive tubular element comprises at least one of a conductive cannula or a conductive coil.

10. The method of claim 1, further comprising: delivering electrical current in a range of 80-120 Watts to the conductive tubular element while the conductive tubular element is cutting into the pseudocyst to create the opening.

11. The method of claim 1, wherein the inner elongate tubular member comprises a body and a wireguide lumen longitudinally extending through the body, the body comprising an insulating material that electrically insulates the wireguide lumen from the conductive tubular element.

12. The method of claim 1, wherein the treatment site is in a gastrointestinal tract of the patient.

13. The method of claim 12, wherein the pseudocyst is adjacent a wall of the gastrointestinal tract, the method further comprising: cutting into the wall before cutting into the pseudocyst in order to create the opening.

14. The method of claim 13, wherein the pseudocyst is adjacent the wall outside the gastrointestinal tract.
